Name ____ Section ________

MTH:2310 DISCRETE MATH
QUIZ 1
DUE: MONDAY FEB. 3

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: (4 points)
Your ability to express your solutions is just as important as mathematical accuracy.
You will be penalized up to 2 points for arithmetic errors.
You will be penalized up to 2 points for disorganized or hard to read solutions.

Problem 1. *(12 points) Propositions*
Determine whether the following statements are propositions. If the statement **is** a proposition, indicate whether it is (T)rue or (F)alse.

a. *(3 points)* "The Milwaukee School of Engineering is located in Salt Lake City, Utah."

b. *(3 points)* "This is a true statement."

c. *(3 points)* "If $x = 5$, then $5 + x = 10$."

d. *(3 points)* "Please write your answer for Problem 1(d) below."

Problem 2. *(12 points) Compound Propositions*
Translate each statement into a compound proposition (using propositional variables and logical operators). Clearly define every variable you use.

a. *(4 points)* "I am early to class or there is traffic on the interstate."

b. *(4 points)* "You did not get an A on the exam and you studied for 10 hours."

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c. *(4 points)* "If I stub my toe, I will cry for at least 2 minutes."

Problem 3. *(6 points) Compound Propositions*
If $p =$ "The sun is extremely hot." and $q =$ "I will get a very serious sunburn today." translate the following compound propositions into words.

a. *(3 points)* $p \wedge q$

b. *(3 points)* $\sim q \leftrightarrow \sim p$

c. *(3 points)* $(p \rightarrow q) \vee q$

Problem 4. *(12 points) Truth Tables*
Fill out a truth table for the following propositional expressions.

a. *(4 points)* $(\sim p \leftrightarrow q) \oplus (q \wedge p)$

b. *(8 points)* $(\sim p \leftrightarrow \sim q) \leftrightarrow (q \leftrightarrow r)$
Hint: A truth table with n distinct propositions must have 2^n rows.